

Acoustic simulation of rooms with boundaries of partially specular reflectivity

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Abstract

A method for calculating the impulse response for rooms with partially specularly reflecting walls is presented. The method combines the extended radiosity method and the conventional methods based on geometrical acoustics, *ray tracing* and *mirror-image methods*, to handle not only specular–specular and diffuse–diffuse reflections, but also diffuse–specular and specular–diffuse ones. © 2001 Elsevier Science Ltd. All rights reserved.

1. Introduction

Computer models, based on geometrical room acoustics, have been developed for a long time [1,2]. In such models, ray tracing and/or mirror-image methods are used, and the wall reflections are considered to be totally specular. In real rooms, a certain portion, d , of the energy is reflected in a diffuse manner while the remaining amount, $1-d$, of the energy is specularly reflected [3], which results in four reflection combinations: diffuse–diffuse, diffuse–specular, specular–specular and specular–diffuse, as shown in Fig. 1. The fraction d is called the diffusion coefficient, which depends on the surface roughness and on the room shape [5,6].

In this paper, a new method is suggested to handle the four reflection combinations described above combining geometrical acoustics and radiant exchange method. The mirror-image method [7] is used to handle the specular–specular while the extended radiosity method [4] deals with the specular–diffuse, diffuse–diffuse and the diffuse–specular reflections.

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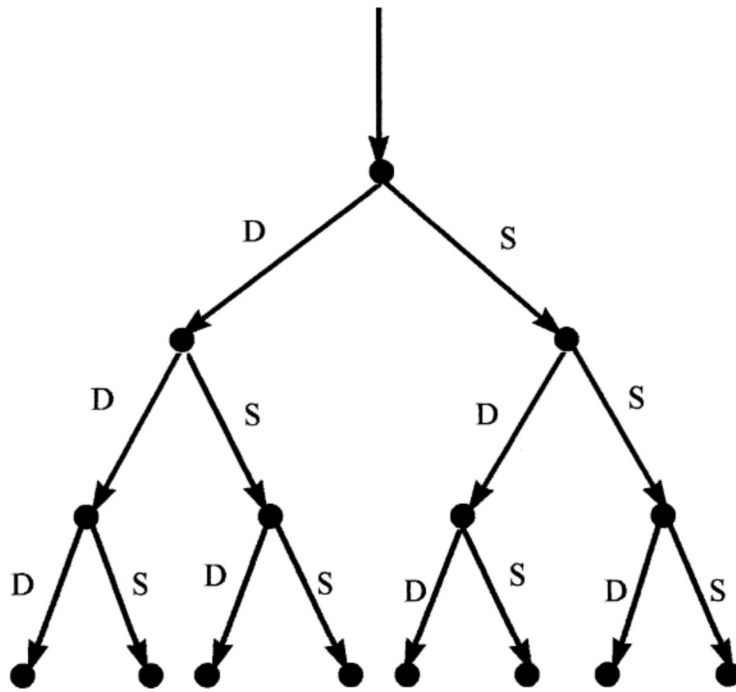


Fig. 1. Illustration of the sound reflections in enclosures: D (diffuse) and S (specular).

2. The extended radiosity method

In the conventional radiosity method, the energy reflected from one element area and directly incident upon another one depends on the form factor between the two elements [8]. The form factor from a certain element i to another element j is defined mathematically as:

$$F_{ij} = \frac{\cos\vartheta_i \cos\vartheta_j}{\pi R_{ij}^2} \Delta A_j \quad (1)$$

where ϑ_i and ϑ_j are the angles between the direction of the reflected energy and the normal to surface element i and j , respectively, ΔA_j is the area of the infinitesimal surface element j and R_{ij} is the distance between the two surface elements, as shown in Fig. 2.

The radiosity method was extended in [4,9] to handle ideal specular reflections. In this method, the energy reflected from one element i and reaching another element j after k specular reflections depends on the form factor between element j and the image of element i at a certain order of reflection, k . $F_{ij,k}$ will refer to this form factor and it is of the same form as that defined in Eq. (1). Fig. 3 shows the radiant energy transfer after one specular reflection.

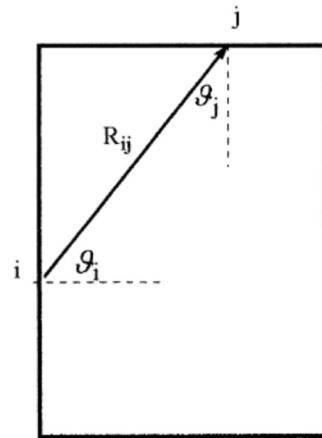


Fig. 2. Radiant exchange in rooms with diffuse reflecting boundaries.

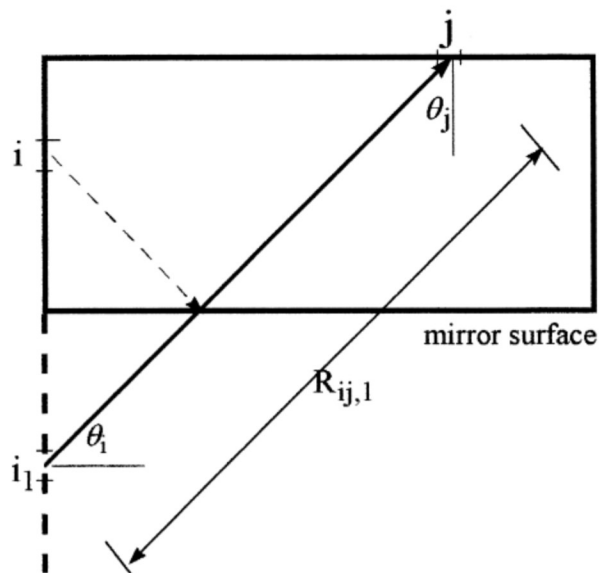


Fig. 3. Radiant exchange in rooms in the presence of specular reflecting boundaries.

This form factor has to be modified to take into account the absorption and the diffusion coefficients of the reflecting surfaces lying in the path between the two elements. It will be called a high-order form factor and this could be derived as follows: The high-order form factor from element i to element j after one specular reflection for a certain path of reflection is

$$HOF_{ij,1}(t) = F_{ij,1} * (1 - \alpha_{s(1)}) * (1 - d_{s(1)}) * \delta\left(t - \frac{R_{ij,1}}{c} + \frac{R_{ij}}{c}\right) \quad (2)$$

where $F_{ij,1}$ is the form factor from a certain mirror image of element i at the first order of reflection and the element j . $\alpha_{s(1)}$ and $d_{s(1)}$ are the absorption and the diffusion coefficient of the reflecting surface. $R_{ij,1}$ is the distance between the image of element i and element j , and R_{ij} is the distance between element i and element j . In general, the high-order form factor from element i and element j after k specular reflections is

$$HOF_{ij,k}(t) = F_{ij,k} * \prod_{id=1}^k ((1 - \alpha_{s(id)}) * (1 - d_{s(id)})) * \delta\left(t - \frac{R_{ij,k}}{c} + \frac{R_{ij}}{c}\right) \quad (3)$$

Taking the summation over all mirror images of element i , that are audible by the receiving element j , the high-order form factor from i to j is given by

$$HOF_{ij}(t) = \sum_{n=1}^N F_{ij,k_n}(n) \delta\left(t - \frac{R_{ij,k_n}(n)}{c} + \frac{R_{ij}}{c}\right) \prod_{id=1}^{k_n} ((1 - \alpha_{s(id)})(1 - d_{s(id)})) \quad (4)$$

where N is the total number of image sources of element i audible by element j , $F_{ij,k_n}(n)$ is the form factor between the n th image source of i at certain order of reflection k_n and the element j , $R_{ij,k_n}(n)$ is the corresponding distance between them, k_n is the diffuse–specular order of reflection of the n th image source, t is the time, c is the sound speed, $s(id)$ is a sequence of k reflecting surfaces connecting i and j , and $\alpha_{s(id)}$ and $d_{s(id)}$ are the corresponding absorption and diffusion coefficients, respectively.

3. The acoustical specular–radiant method

In this method, all boundaries are divided into a number of very small areas ΔA . Independently of the receiving position, the sound energy received by each element is calculated. Each element receives a totally specular amount of energy from the source, and another mixed specular and diffuse amount due to the contribution from all other elements. The purely specular part of the element response is calculated by applying the mirror-image method. The sound energy emitted from the source and incident directly up on a certain surface element, and that received by the surface element after a certain number of specular reflections could be calculated from the following equation

$$B_i(t) = \frac{W}{\Delta A_i} \sum_{n=0}^N \left(F_{si,r(n)} \delta\left(t - \frac{R_{si,r(n)}}{c}\right) \right) \quad (5)$$

where W is the total acoustical power emitted by the source. N is the total number of image sources, that are audible to the surface element i . $r(n)$ is the order of reflection of the n th source. $F_{si,r(n)}$ is the form factor between an elementary area and a certain image source, which is audible to that element which could be calculated using the Eq. (6), taking into account the acoustical characteristics of the reflecting boundaries.

$$F_{si,r(n)} = \frac{\cos\theta_{i,r(n)}}{4\pi R_{si,r(n)}^2} \Delta A_i \prod_{id=1}^{r(n)} ((1 - \alpha_{s(id)})(1 - d_{s(id)})) \quad (6)$$

where $\theta_{i,r(n)}$ is the angle between the line connecting the element i to the image source at a certain order r and the normal to the element area. $R_{si,r(n)}$ is the distance between the element i and the n th image source at certain order of reflection r . $\alpha_{s(id)}$ and $d_{s(id)}$ are the absorption and the diffusion coefficients of the in-between reflecting surfaces, respectively. $n = 0$ corresponds to the direct energy received from the sound source, i.e. $r = 0$ and

$$F_{si,0} = \frac{\cos\theta_i}{4\pi R_{si}^2} \Delta A_i \quad (7)$$

where R_{si} is the direct distance between the sound source and the element i .

This purely specular part will be used to find the part of the response resulting from mixed specular and diffuse reflections. This later part is obtained by considering the sound-energy transfer between different elements, which depends on the position of the receiving element with respect to the source element, that could be summarised in the following two cases:

- *Case 1:* The energy reflected directly to a certain element depends on the form factor between the two elements as defined by Eq. (1). The time delay is calculated by dividing the distance between the two elements by the sound speed.
- *Case 2:* The energy reflected to a certain element, and reaching it after k specular reflections depends on the high-order form factor, which is defined by Eq. (4).

Fig. 4 illustrates the energy transferred from surface element i to surface element j . In general, the sound energy reaching element j due to the contribution from other elements could be written in the following form:

$$B_j(t) = \sum_{\substack{i=1 \\ i \neq j}}^M \left(B_i \left(t - \frac{R_{ij}}{c} \right) \rho_i d_i \otimes (F_{ji} V_{ij} \delta(t) + HOF_{ji}(t)) \right) \quad (8)$$

where B is the sound energy per unit area per s, M is the total number of surface elements, and V_{ij} is the visibility between the elements i and j :

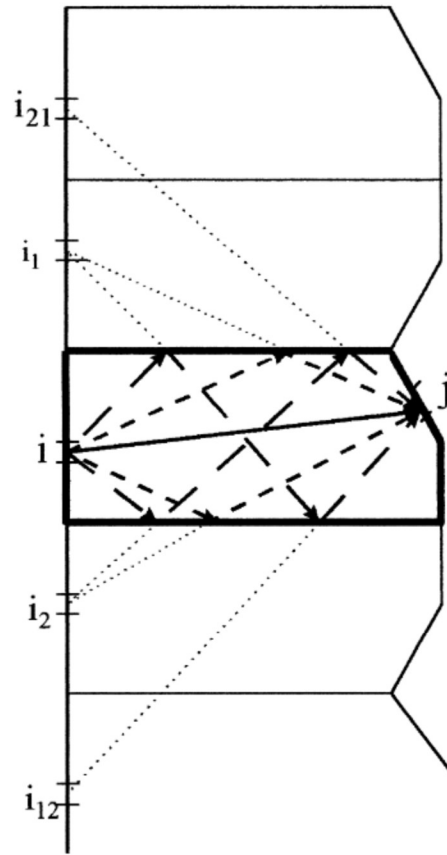


Fig. 4. Sound energy transfer from element i to element j . Solid line: sound reaching j directly; dotted lines: sound reaching j after k reflections ($k = 1, 2$).

$$V_{ij} = \begin{cases} 1 & \text{element } i \text{ is visible from element } j \\ 0 & \text{element } i \text{ is hidden to element } j \end{cases} \quad (9)$$

The symbol $\otimes$ denotes convolution. The term $F_{ji} V_{ij} \delta(t)$ refers to the energy reflected from i and incident directly upon j , whereas the term $HOF_{ji}(t)$ refers to the energy reflected from i and received by j after a certain number of specular reflections.

Briefly, Eq. (8) is used to simulate the paths of mixed diffuse–specular reflections and those of totally diffuse reflections, whereas Eq. (5) is used to find the direct sound and the paths of totally specular reflections. The energy response of each surface element is calculated independently from the receiving position. At a certain receiving position, the purely specular part of the response is calculated by applying the mirror-image method, and the corresponding sound intensity due to each mirror-image is calculated using the following equation:

$$I_n(t) = \frac{W}{4\pi R_n^2} \prod_{id=1}^{r(n)} ((1 - d_{s(id)})(1 - \alpha_{s(id)})) \delta\left(t - \frac{R_n}{c}\right) \quad (10)$$

where R_n is the distance between the n th mirror image source and the receiving position. $r(n)$ is its order of reflection. The summation is taken over all mirror-images that are audible to the receiver to obtain the purely specular part of the response. To predict the sound intensity in the part of the response that contains mixed diffuse and specular reflections, each element is considered as a source and the energy transfer from this source to the receiver depends on their relative position with respect to each other (Fig. 5). Two types of energy may be received from each element. One is that reflected from the element and is directly incident upon the receiver, and this is expressed as

$$I_D(t) = B_j \left(t - \frac{R_{j,\text{rec}}}{c}\right) d_j (1 - \alpha_j) \Delta A_j V_{\text{rec},j} \frac{\cos \theta_j}{\pi R_{j,\text{rec}}^2} \quad (11)$$

where j denotes a certain element number. θ_j is the angle between the normal to the element area and the line connecting the element to the receiver. $R_{j,\text{rec}}$ is the distance from the element to the receiver. $V_{\text{rec},j}$ is the visibility factor between a certain element, j , and the receiver, and it is defined in Eq. (9).

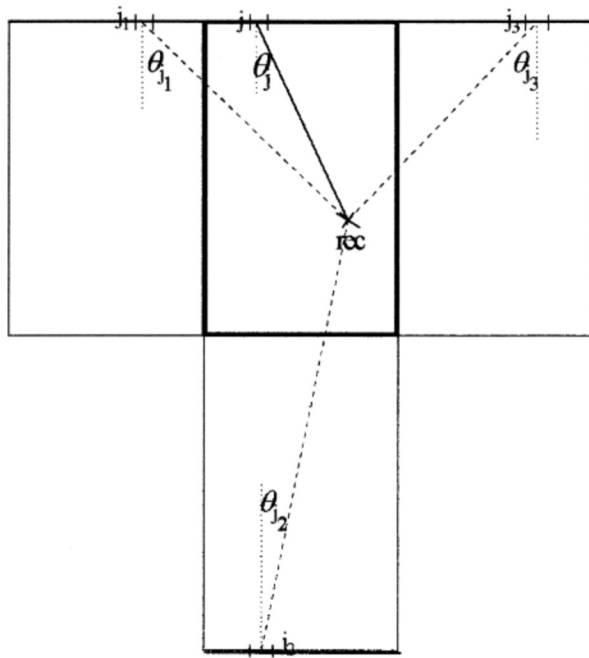


Fig. 5. Mixed diffuse and specular reflections received from element j .

The second type of energy is that reflected from a certain element and reaches the receiver after a certain number of specular reflections, and this could be mathematically written as

$$I_{S,D}(t) = B_j \left(t - \frac{R_{j,\text{rec}}}{c} \right) d_j (1 - \alpha_j) \Delta A_j \otimes \frac{F(\text{rec}, j)}{\pi} \quad (12)$$

where $F(\text{rec}, j)$ is a function depending on the relative position of the receiver with respect to the element and on the acoustical characteristics of the reflecting surfaces, that form the path from the element to the receiver, and it is defined as

$$F(\text{rec}, j) = \sum_{n=1}^N \left(\frac{\cos \theta_{j_n}}{R_{j_n, \text{rec}}^2} \delta \left(t - \frac{R_{j_n, \text{rec}}}{c} + \frac{R_{j, \text{rec}}}{c} \right) \prod_{id=1}^{k_n} ((1 - d_{s(id)})(1 - \alpha_{s(id)})) \right) \quad (13)$$

where N is the total number of the mirrored images of element j that are audible by the receiver. θ_{j_n} is the angle between the normal to the mirror-image of a certain surface element, j , and the line connecting the receiver and that mirror-image. k_n is the order of reflection of the n th mirror-image of that element. The sum of the energies, given by Eqs. (11) and (12), represents the energy received from a certain element after suffering from mixed diffuse and specular reflections. Taking the summation over all elements forming the enclosure, the sound intensity at the receiver is

$$I_{ns}(t) = \sum_{\forall j} (I_D(t) + I_{S,D}(t)) \quad (14)$$

Eq. (14) simulates the sound energy received due to mixed diffuse and specular reflections and that due to a series of totally diffuse reflection, whereas using Eq. (10) the sound energy received after a series of totally specular reflection is calculated. Note that the part of the element response resulting from specular reflections only will not be specularly received. In this case, Eq. (11) will only be used, i.e. the sound energy of a certain element, that resulted from a totally specular reflections will be received in a diffuse manner.

4. Room simulation

In order to minimise the required storage and to speed up the simulation, a limit of the diffuse–specular reflection order (see Section 3) has to be set, such that the energy reaching certain element after a number of specular reflections higher than that limit could be neglected. To determine that limit, a room of dimensions $20 \times 30 \times 10 \text{ m}^3$ was simulated, the absorption coefficient was set to 0.2 and the diffuse coefficient to 0.4. The fraction of the diffuse energy that is reflected specularly from a

certain element to all other elements was calculated at different order of reflections, the result is normalised and plotted in Fig. 6. It shows that the biggest portion of diffuse energy, that is reflected specularly, takes place at the first and the second orders. This fact allows k_n to be limited to 2.

The previous room was simulated. The absorption coefficients of the walls are shown in Table 1. The diffusion coefficient is considered to be uniformly distributed over all walls. For a non-directive source position of (2.5,15,3) in m, the room

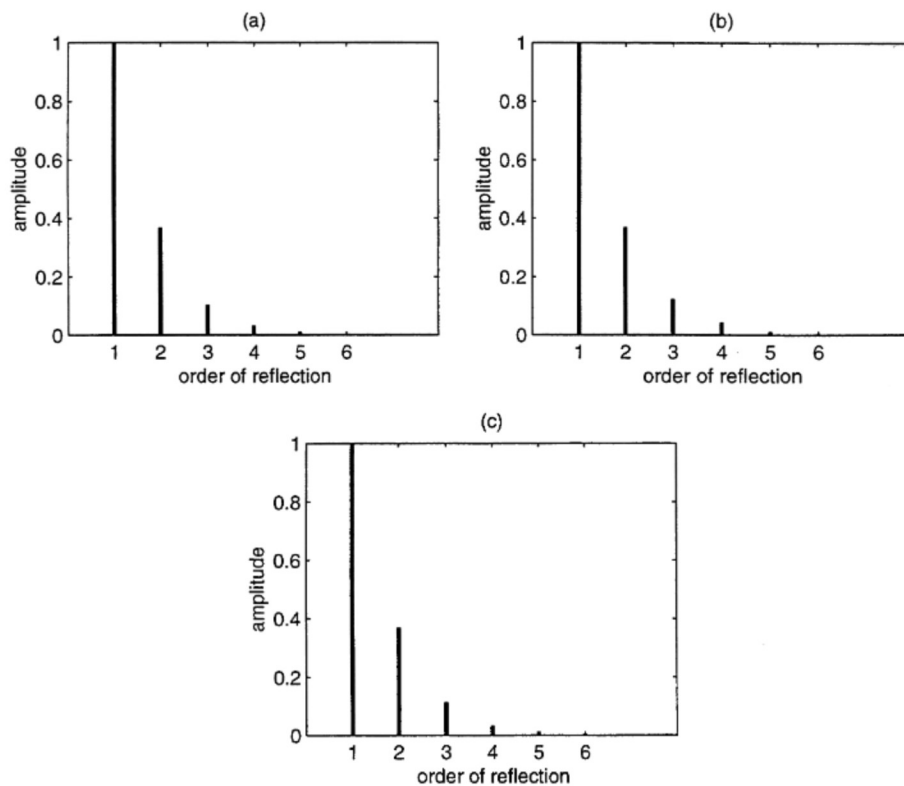


Fig. 6. Normalised fraction of diffuse energy reflected specularly at different order of reflection from an element positioned: (a) at a corner; (b) on an edge; (c) at the centre of a surface.

Table 1

The absorption coefficients assigned to each boundary of the simulated room

Floor	0.9
Ceiling	0.1
Long side walls	0.3
Short side walls	0.6

impulse response at a receiving position of (15,10,4) in m was calculated. The simulation is repeated assigning different values of the diffusion coefficient to the walls. Figs. 7 and 8 show the magnitude of the calculated sound pressure and the corresponding decay curve for diffusion coefficient equal to 0.0 and 0.6, respectively. It is shown that the impulse response, that was calculated taking into account diffuse reflection, is more dense, which yields a smoother decay curve. The reverberation time is calculated from the decay curve, and it is compared to that calculated after Sabine's and Eyring's equations as shown in Table 2. A more exactly reverberation time of enclosures with diffusely reflecting boundaries was calculated by Kuttruff by means of an iteration scheme, that is based on the integral equation. In [3], some results were presented. One arbitrary room example is chosen: a room of dimensions $30 \times 20 \times 10$ m, the floor (20×30 m) is considered to be a perfect absorber ($\alpha = 1$),

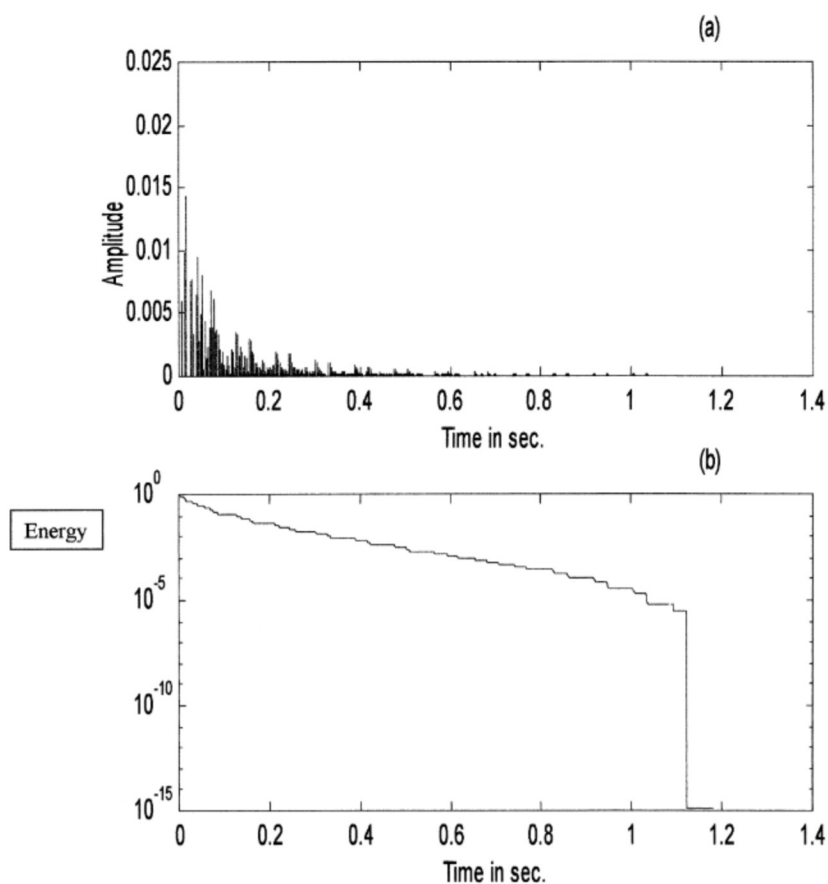


Fig. 7. Room size: $20 \times 30 \times 10$ m, diffusion coefficient = 0.0: (a) magnitude of the sound pressure; (b) the corresponding decay curve.

while the remaining walls are considered to have a zero absorption coefficient. Assigning a uniform diffusion coefficient of 1.0 to the walls, the room is simulated using the specular–radiant model, and the reverberation time is estimated and is compared with that calculated after Kuttruff. The results are presented in Table 3. It is clear that the reverberation time calculated using the specular–radiant model is nearly equal to that calculated after Kuttruff, noting that Sabine’s and Eyring’s equations do not take into account the room shape.

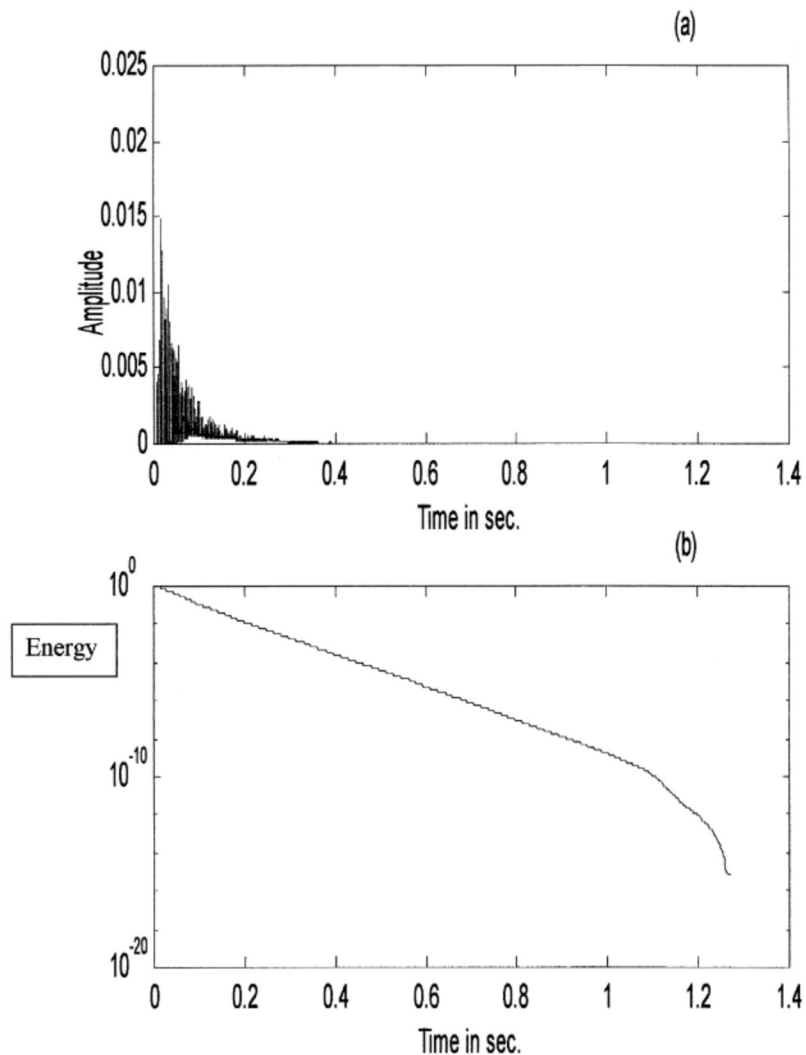


Fig. 8. Room size: 20×30×10 m, diffusion coefficient = 0.6: (a) magnitude of the sound pressure; (b) the corresponding decay curve.

Table 2

The reverberation time (s) of a room of dimensions $30 \times 20 \times 10$ m; the absorption coefficients are shown in Table (1)

The used method		RT
Sabine		0.906
Eyring		0.714
Specular–radiant method in case of uniform diffusion coefficient of	0	1.13
	0.2	0.978
	0.4	0.756
	0.6	0.684
	0.8	0.693
	1.0	0.72

Table 3

Comparison of reverberation time for a room of dimensions $30 \times 20 \times 10$ m; the floor (20×30 m) is a perfect absorber, and the remaining walls have 0.0 absorption

The used method		RT (s)
Sabine		1.63
Eyring		1.4
Kuttruff		1.22
Specular–radiant method, diffusion coefficient = 1.0		1.09

5. Conclusion

In practical situations, it is reasonable to consider the reflections from the walls to be partially specular. A computer model have been suggested to handle all type of reflections described above. The method allows to calculate the wall responses independently from the receiving position. Then at a certain receiving position the room impulse response could be obtained. The program run time is in a reasonable range and it depends on the number of wall elements.

References

- [1] Lehnert H, Blauert J. Principles of binaural room simulation. *Applied Acoustics* 1992;36:259–91.
- [2] Vian JP, Martin J. Binaural room simulation: practical uses and applications. *Applied Acoustics* 1992;36:293–305.
- [3] Kuttruff H. Energetic sound propagation in rooms. *Acustica — Acta Acoustics* 1997;83:622–8.
- [4] Rushmeier HE. Extending the radiosity method to transmitting and specularly reflecting surfaces. MSc thesis, Faculty of the Graduate School of Cornell University, January 1986.
- [5] Lam YM. The dependence of diffusion parameters in a room acoustics prediction model on auditorium sizes and shapes. *Journal of the Acoustical Society of America* 1996;100(4):2193–203.
- [6] Vorländer M, Schaufelberger T. Messung des Diffusitätsgrades rauher Fläachen. *Fortschritte der Akustik; DAGA'90*. p. 827–30.

- [7] Borish J. Extension of the image model to arbitrary polyhedra. *Journal of the Acoustical Society of America* 1984;75(6):1827–36.
- [8] Siegel R, Howell JR. Thermal radiation heat transfer. Hemisphere Publishing Corporation/McGraw-Hill Book Company, 1981.
- [9] Eckert ERG, Sparrow EM. Radiative heat exchange between surfaces with specular reflection. *International Journal of Heat and Mass Transfer* 1961;3:42–54.